Inline Optics Standard Operating Procedure

Version 1.0

Authors

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Protocol Revision Record

Version	Date Implemented	Description of Revisions
1.0	10 May 2025	Initial release in format standardized for Bio-GO-SHIP

Acronyms, Abbreviations, Terminology

Term	Definition
Inline/underway water	Sampling water that is pumped from beneath the ship to a receiving spigot in an onboard laboratory
DI water	Deionized water from the Milli-Q system
BB-3	WetLabs sensor measuring scattering of particles
AC-S	WetLabs sensor measuring the spectra of absorption and beam attenuation of particles
FRRF	Fast repetition rate fluorometer
Data junction box	Also called the power distribution center- a box that contains power cable connections for all parts of the inline optical system
GO-SHIP	Global Ocean Ship-Based Hydrographic Investigations Program
Bio-GO-SHIP	Biological GO-SHIP

Background

Summary

This protocol describes the continuous sampling of optical properties of surface water while onboard a research vessel. Notes for the beam transmissometer attached to the CTD rosette are also included.

Method description and rationale

The inline system draws water in from below the ship (~5 m) using a diaphragm pump and is directed through a debubbler before passing through the inline set-up. The inline system will vary depending on cruise- typically including a WetLabs BB-3 in a custom dark casket, WetLabs AC-S, or a fast repetition rate fluorometer (FRRF), as well as a flow meter. The diaphragm pump and debubbler are pertinent to preserve the physiological characteristics of particles and cells before they reach the instrumentation. The system is designed to continuously record optical data from surface waters, as well as record flow rate of sampling water and live GPS coordinate timestamps.

Spatial coverage and environment(s) of relevance

This protocol can be used across any marine environment to effectively collect continuous optical data from surface waters.

Ship space requirements

Access to a sink with inline water supplied by a diaphragm or positive displacement pump. Sink outflow must be directed overboard as to not fill holding tanks. Optical instruments will be secured in the sink, with nearby counterspace needed for data junction box and laptop (~2 square feet of counter space).

Personnel Required

One person that is trained in instrument operation and maintenance. Typically, two individuals that operate on 12-hour shifts.

Safety

There are no major safety concerns with this protocol. Take care when cleaning instruments as to not shatter sensor faces or fragile AC-S tubes.

Time needed to execute the procedure

Instruments continuously collect data. Cleaning frequency and time spent cleaning instruments is dependent on the needs of the cruise.

Standard Operating Procedure

Prior to ship's departure, the inline system (including: debubbler, tubing and connectors to instrumentation, filter, flow meter, data junction box) should be set up by trained personnel.

Once the cruise is in transit, the diaphragm pump must be cleaned with bleach.

- Each cruise's engineer personnel will have different ways of managing this. They can pour liquid bleach into the system and allow to sit for at least one hour, or they may utilize bleach tablets and allow the system to flush for several hours continuously while the bleach tablet dissolves.
- Once bleaching is complete, the system must be continuously flushed for at least one hour to remove any remaining bleach from the system.

When instruments are plugged in and connected:

- Be sure power (lights) are going to the data/power distribution center. If not, check power cables.
- Start one session of Inlinino (via Terminal or double-clicking icon) and connect to one instrument. The names of instruments may vary per cruise and separate instructions will be supplied as needed.
- Choose WetLabs BB3 499 from drop down menu and press "Load"
- Select the "Open" button where it says Serial Port.
- Choose the correct port for this instrument (see tape on laptop) and then open.
    - At this point data should start streaming into the display.
- Open a second session of Inlinino via Terminal and connect to the AC-S 111 and follow similar steps above (see tape on laptop on how to use Terminal to open a second session).
- Open another session of Ininino and connect to the NMEA data stream and see that GPS data is streaming.
- Open another Inlinino session and open the flow rate and switch control panel. Check that the flow rate is being collected. Test that the switch control is working and set an appropriate schedule for the filtered/no-filtered time periods.

- If data looks “stable” and no sign of bubbles/spike you can begin logging data for each instrument, flow rate, and GPS data streams by pressing start on each Inlinino window (4-5 windows total depending on the number of instruments)

Filtration for inline system should be set for 10 minutes every hour at the top of the hour. This provides the “seawater blank” (water + dissolved constituents) for each instrument.

If the data does not reach a minimum in that 10-minute period, you may need to increase the filtration period to 15 minutes. Low flow rates in the inline system will slow the clearing of sweater from the tubing and BB3 casket.

Cleaning: clean each instrument once per week. However, if you notice a trend in the filtered value of the BB3 (in the casket) it may need more frequent cleaning. Try to keep track of that filtered value a couple times per day. Before cleaning, stop data collection and remove tubing going into the first instrument such that it can drain into the sink.

BB3: For the BB3 and casket, drain the casket and open the side panel. Rinse the inside with freshwater and DI. You can use a sponge or paper towels to initially wipe the casket sides (but not the lens of the bb3). Wipe the instrument lens and inside surface of the casket with pre-moistened lens wipes and rinse with DI before putting side panel back onto the casket. Be careful not to lose the o-ring on the casket that seals the side panel.

AC-S: To clean the AC-S, carefully remove the a and c tubes. These should be labelled so you know which one to put back on which side after cleaning. Rinse each tube with DI. Carefully push a pre-moistened lens wipe through each tube, likely multiple time or with multiple wipes, to ensure the wipe comes out clean and when looking through the tubes up towards the overhead lights it looks clean and “smooth”. Use a lens wipe to clean the sensor windows and rinse with DI.

Data: Data from each instrument, flow meter, and GPS are being stored in respective folders on the laptop’s desktop. Once per week, back up this data to the cruise’s shared drive. Keep a notes app file to record time and date of technical issues, cleanings, data back-ups, beam transmissometer info, etc.

Beam transmissometer: The beam transmissometer on the CTD Rosette needs to be cleaned daily. Rinse windows with DI, wipe with lens wipe with isopropyl or pre-moistened lens wipe, and rinse with DI again.

Once per week perform a dark cast by putting electrical tape on the windows of the transmissometer during a standard bio-cast and then remove the tape and clean the windows before the next cast. These “dark casts” are used for calibrating temperature and pressure impacts on the data.

Basic troubleshooting

Contact Jason Graff via WhatsApp for any issues encountered.